

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems.(L4,L5)

Assessment Tool : Alternative Solution Design
Weightage: 20%

QUESTIONS:

Total Marks: 25

1. A company needs to develop a web service for a mobile and web application. Analyze both **REST and SOAP** approaches and design an appropriate solution by selecting one. Justify your choice based on performance, scalability, and security requirements.
2. An online platform expects a large number of users accessing its services simultaneously. Design a **scalable service architecture** showing how client, server, API, and database interact. Propose an alternative design to improve scalability and justify your approach.
3. A banking application requires secure communication between client and server. Analyze available communication protocols and **select the most suitable protocol**. Justify your selection based on security, reliability, and performance.
4. An application needs to integrate with third-party services such as payment gateways or maps. Design an **API integration strategy**, including request handling, authentication, and response processing. Suggest an alternative approach to improve efficiency or reliability.
5. A web service frequently encounters errors during client requests. Design an **error handling and fault tolerance mechanism**. Propose alternative strategies to improve system reliability and ensure smooth user experience.

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly defines problem, objectives, and all requirements accurately	Understands problem with minor gaps	Partial understanding; misses key requirements	Misinterprets or fails to understand problem
Generation of Solutions	25%	Develops multiple innovative and feasible alternatives with clear differentiation	Generates relevant alternatives with minor limitations	Limited or repetitive alternatives	Fails to generate meaningful alternatives
Evaluation of Alternatives	25%	Critically compares alternatives using appropriate criteria and metrics	Compares alternatives with some justification	Basic or superficial comparison	No proper comparison conducted
Solution Selection	20%	Selects best solution with strong justification and decision rationale	Selects suitable solution with moderate justification	Selection is weakly justified	No clear or justified selection
Feasibility	10%	Applies relevant concepts ensuring high feasibility and practicality	Applies concepts with minor issues	Limited feasibility or incorrect application	Solution is impractical or conceptually incorrect